
Volatility Trading with Delta-Hedged ATM Straddles

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Abstract

Abstract to be completed in the final version.

1 Introduction

This report studies a low-frequency volatility trading strategy implemented through one-month at-the-money (ATM) straddles on U.S. equities and exchange-traded funds. The core idea is simple: options embed the market’s forward-looking assessment of risk through implied volatility, while the underlying asset reveals realized volatility only after price moves occur. A persistent gap between the two creates a natural trading signal. When implied volatility appears rich relative to an appropriate benchmark, the strategy sells volatility through a short straddle; when implied volatility appears cheap, it buys volatility through a long straddle.

The key challenge is that an option position is not a pure volatility bet. A straddle carries directional sensitivity, convexity, time decay, and sensitivity to changes in interest rates and implied volatility itself. For that reason, the strategy overlays delta hedging in the underlying and evaluates performance through a daily Greeks-based attribution. The objective is not merely to report returns, but to show whether profits arise from the intended volatility exposure rather than from incidental equity direction.

The intended audience is assumed to be familiar with stocks, returns, and standard backtesting ideas, but not necessarily with option trading. The discussion therefore introduces the economic intuition of a straddle, explains the role of implied volatility and Greeks, and then presents the trading rules and empirical results in a self-contained way.

2 Options Preliminaries

2.1 Call, put, and straddle

A call option gives its holder the right, but not the obligation, to buy the underlying asset at a predetermined strike price K before or at expiry. A put option gives the right to sell at the same strike. The holder of a long option pays the option price upfront and has limited downside equal to that payment. The seller of the option receives that payment upfront but assumes the contingent liability.

At expiry T , the terminal payoffs of a call and a put are

$$\text{Call payoff} = \max(S_T - K, 0), \quad \text{Put payoff} = \max(K - S_T, 0),$$

where S_T is the underlying price at expiry.

The strategy trades an at-the-money straddle, which is the combination of one call and one put written on the same underlying, with the same strike and the same expiry. At expiry, the payoff of a long straddle is

$$\max(S_T - K, 0) + \max(K - S_T, 0) = |S_T - K|,$$

so the position benefits from sufficiently large moves in either direction. This makes the straddle the natural vehicle for expressing a view on volatility rather than on market direction.

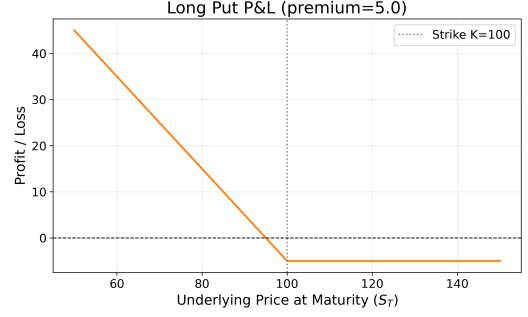
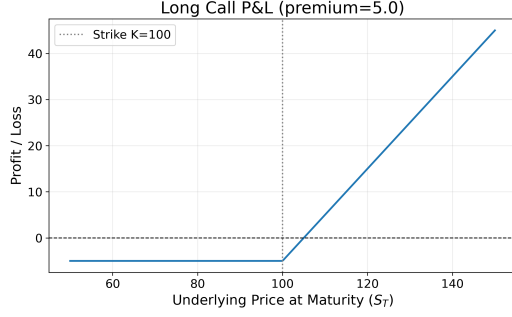


Figure 1: Illustrative maturity P&L of a long call and a long put.

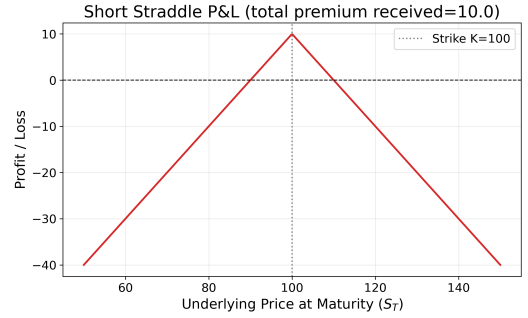
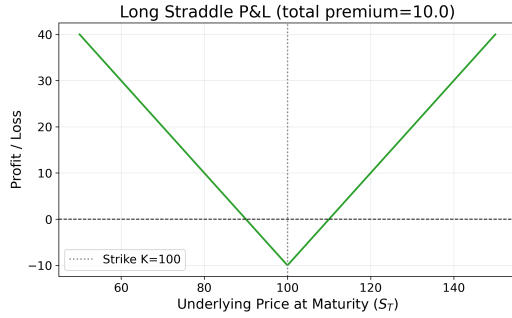


Figure 2: Illustrative maturity P&L of long and short ATM straddles.

2.2 Black–Scholes model

This report uses the Black–Scholes model as the starting point for option valuation. The model provides a tractable mapping between an option's value and the key state variables that drive it: the underlying price, strike, time to maturity, risk-free rate, and volatility. Although listed equity options are American-style in legal form, Black–Scholes remains a standard and useful approximation for short-dated liquid contracts, especially when the main objective is to infer implied volatility and measure risk exposures consistently.

For a European call and put with time to maturity $\tau = T - t$, risk-free rate r , and volatility σ , prices are

$$C = S_t N(d_1) - K e^{-r\tau} N(d_2), \quad P = K e^{-r\tau} N(-d_2) - S_t N(-d_1),$$

where

$$d_1 = \frac{\ln(S_t/K) + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \quad d_2 = d_1 - \sigma\sqrt{\tau},$$

and $N(\cdot)$ denotes the standard normal cumulative distribution function.

2.3 Greeks

The value of an option depends on the underlying price, implied volatility, time to expiry, and interest rates. The Greeks summarize local sensitivities:

$$\Delta = \frac{\partial V}{\partial S}, \quad \Gamma = \frac{\partial^2 V}{\partial S^2}, \quad \Theta = \frac{\partial V}{\partial t}, \quad \nu = \frac{\partial V}{\partial \sigma}, \quad \rho = \frac{\partial V}{\partial r},$$

$$\text{Vanna} = \frac{\partial^2 V}{\partial S \partial \sigma}, \quad \text{Volga} = \frac{\partial^2 V}{\partial \sigma^2}.$$

For the present strategy, the most important interpretations are straightforward. Delta measures first-order directional exposure to the underlying. Gamma captures convexity in the underlying price. Vega measures sensitivity to changes in implied volatility. Theta represents time decay. Vanna and volga become relevant when the underlying and implied volatility move jointly or when volatility changes materially.

A second-order expansion of option value over a short interval provides the basic P&L map:

$$dV \approx \Delta dS + \frac{1}{2}\Gamma dS^2 + \nu d\sigma + \Theta dt + \rho dr + \text{Vanna} dS d\sigma + \frac{1}{2}\text{Volga} (d\sigma)^2 + \varepsilon,$$

where ε captures higher-order terms. This decomposition follows the standard treatment of option risk attribution in Bossu [1]. It is useful later when interpreting whether a delta-hedged straddle is actually earning volatility-related P&L.

2.4 Implied volatility

Implied volatility is the value of σ that makes the Black–Scholes price match the observed market price of an option or option package. In that sense, implied volatility is not directly observed; it is backed out from price. Practitioners often quote options in implied-volatility terms because volatility provides a standardized way to compare option richness across time and across contracts.

This interpretation is central to the strategy studied in this report. The trading decision is not driven by a view on whether the stock will rise or fall. Instead, it is driven by whether the option market appears to be charging too much or too little for future uncertainty relative to a benchmark for expected or normal volatility.

3 Methodology

3.1 Data

The empirical sample consists of three exchange-traded funds, namely SPY, QQQ, and DIA, together with a set of individual U.S. equities: AAPL, ACMR, AMD, GOOG, INTC, META, and TSLA. These assets provide a useful mix of broad market exposure and single-name volatility, allowing the strategy to be examined across instruments with different liquidity conditions, sector compositions, and event risk. The individual stocks are not selected through a formal screening rule; rather, they are chosen largely at random from names with which I am personally familiar, so the single-stock sample should be interpreted as illustrative rather than exhaustive.

Although the strategy is implemented on all of these symbols, this report focuses primarily on the performance of SPY. The reason is practical as well as economic: SPY is the most popular and most liquid ETF in the sample, its options market is particularly deep, and its prices are therefore less affected by idiosyncratic microstructure noise than those of many single stocks. For that reason, SPY provides the cleanest setting for evaluating whether the strategy behaves as intended.

For each symbol and each month, the strategy selects an at-the-money straddle associated with the next monthly expiry. The strike is fixed at entry and the position is monitored through the month. As the underlying moves, the contract drifts away from exact at-the-money status; this is a deliberate simplification consistent with a low-frequency monthly trading design.

3.1.1 Data construction details

Each daily observation combines the underlying close, cash dividends, a short-dated risk-free rate, and the closing prices of the selected call and put. From these inputs, the dataset computes realized volatility, leg-level implied volatilities, a single straddle implied volatility that matches the observed straddle price, and the standard Greeks for both individual legs and the straddle as a whole. All volatility measures are annualized using a 252-trading-day convention.

Positions are delta hedged using the underlying. If the absolute net delta of the portfolio exceeds the re-hedging threshold, the stock hedge is adjusted and hedge shares are rounded to integers. This overlay does not remove all directional risk, but it reduces first-order exposure and makes the strategy more interpretable as a volatility trade rather than a disguised directional position.

3.2 Trading signal

All four agents share the same execution layer. A long-volatility position corresponds to buying one straddle, while a short-volatility position corresponds to selling one straddle when shorting is allowed. Positions are closed when the signal reverts toward its neutral region, and the hedge is updated whenever the net delta drifts beyond the chosen threshold.

3.2.1 Realized volatility benchmark

The classic benchmark in volatility trading is the volatility risk premium,

$$\text{VRP}_t = \text{IV}_t - \text{RV}_t.$$

Here, IV_t denotes the implied volatility of the traded straddle and RV_t denotes realized volatility estimated from recent underlying returns. The intuition is direct: if implied volatility is high relative to realized volatility, the option market may be overpricing future uncertainty; if implied volatility is low relative to realized volatility, options may be relatively cheap.

Realized volatility is computed from a rolling 20-day window of daily log returns. If $r_t = \ln(S_t/S_{t-1})$, then

$$\text{RV}_{t,20} = \sqrt{252} \sqrt{\frac{1}{20} \sum_{i=1}^{20} (r_{t+1-i} - \bar{r}_t)^2},$$

where $\bar{r}_t$ is the mean return over the same window.

This specification is the most familiar and transparent benchmark in the report, because it compares option-implied uncertainty with the variability recently observed in the underlying market.

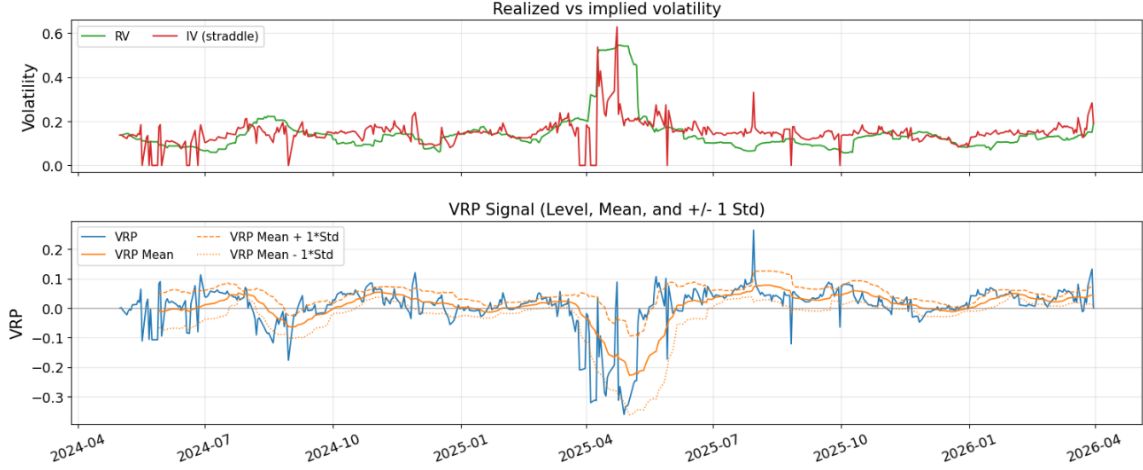


Figure 3: SPY implied volatility and realized volatility in the upper panel, and the corresponding volatility risk premium in the lower panel. The lower panel also reports the rolling mean together with mean ± 1 standard deviation bands.

3.2.2 GARCH-based benchmark

The second benchmark addresses an important mismatch in a plain VRP comparison. Implied volatility reflects the market's expectation of volatility from today to option expiry, while realized volatility is computed from historical returns and does not guarantee persistence into the future. In other words, VRP compares a backward-looking statistic with a forward-looking market expectation.

To make the comparison more consistent in horizon, this benchmark replaces backward-looking realized volatility with a model-based forecast from a GJR-GARCH(1,1) specification with Student- t innovations. This choice allows for volatility clustering and asymmetric responses to negative returns, both of which are common in equity data. The model is estimated on a trailing window, refit periodically, and projected forward over the remaining trading days to the option expiry.

The corresponding spread is

$$IV_t - \widehat{RV}_t^{\text{GARCH}}.$$

This benchmark is designed to compare current option-implied volatility with a forward-looking conditional volatility estimate rather than with a purely historical measure. Operationally, the signal asks whether market-implied future volatility is aligned with model-predicted future volatility.

3.2.3 Long-horizon realized-volatility benchmark

A third benchmark compares current straddle implied volatility with a slow-moving historical average of realized volatility. Specifically, it uses the mean of the prior 60 observations of daily realized volatility as a long-horizon anchor. The corresponding spread is therefore

$$IV_t - \overline{RV}_t^{\text{long}},$$

where $\overline{RV}_t^{\text{long}}$ is computed from past values only.

The economic logic is a convergence chain. Near expiry, option-implied volatility should converge to instantaneous realized volatility. A second empirical observation is that realized volatility itself tends to mean-revert. Combining these ideas gives a practical hierarchy: IV converges to RV, and

RV converges to its long-horizon mean; therefore IV is expected to gravitate toward long-horizon RV as expiry approaches.

This benchmark differs conceptually from the first two. It does not compare implied volatility with the most recent realized-volatility estimate or with a model-based forecast for option expiry. Instead, it compares implied volatility with a slow-moving historical level that realized volatility itself tends to revisit over time.

3.2.4 Z-score normalization

The trading logic compares current straddle implied volatility with these benchmarks through a rolling 20-day z-score,

$$Z_t(X) = \frac{X_t - \mu_{t,20}(X)}{\sigma_{t,20}(X)},$$

where X_t denotes the spread of interest and $\mu_{t,20}(X)$ and $\sigma_{t,20}(X)$ are its rolling mean and standard deviation. A sufficiently negative z-score indicates that options appear cheap relative to the benchmark, while a sufficiently positive z-score indicates that options appear expensive. This normalization makes the trading threshold comparable across different volatility regimes.

3.2.5 Signal summary

The implementation includes four agents built on these benchmarks. The hard-threshold rule and percentile rule both use the classic VRP idea, but they differ in how entry thresholds are defined. The GARCH-based rule replaces realized volatility with a conditional forecast. The long-horizon rule compares current implied volatility with a slow-moving historical mean of realized volatility.

Table 1: Summary of trading rules.

Agent	Long-volatility entry	Short-volatility entry	Exit rule
Hard-threshold VRP	$Z_t(\text{IV} - \text{RV}) < -k$	$Z_t(\text{IV} - \text{RV}) > k$	z-score crosses zero
Percentile VRP	rolling $Z_t(\text{IV} - \text{RV})$ in lower empirical tail	rolling $Z_t(\text{IV} - \text{RV})$ in upper empirical tail	z-score crosses historical median
GARCH spread	$Z_t(\text{IV} - \widehat{\text{RV}}^{\text{GARCH}}) < -c$	$Z_t(\text{IV} - \widehat{\text{RV}}^{\text{GARCH}}) > c$	z-score crosses zero
Long-horizon spread	$Z_t(\text{IV} - \overline{\text{RV}}^{\text{long}}) < -c$	$Z_t(\text{IV} - \overline{\text{RV}}^{\text{long}}) > c$	z-score crosses zero

The reported implementation uses the parameter settings summarized below.

Table 2: Shared execution parameters.

Parameter	Description	Value
<code>allow_short</code>	Whether short-volatility positions are allowed	True
<code>delta_hedge</code>	Whether the straddle is hedged with the underlying	True
<code>rehedge_threshold</code>	Absolute net delta that triggers hedge rebalancing	0.05

Table 3: Agent-specific signal parameters.

Agent	Parameter	Value
Hard-threshold VRP	k	1.0
Percentile VRP	Entry percentile	0.20
GARCH spread	Entry threshold c	1.0
Long-horizon IV spread	Entry threshold c	1.0
Long-horizon IV spread	Long-term RV window	60

All parameters are fixed before the backtest is run. They are not fitted, trained, or optimized on the in-sample period, and the same group of parameter values is applied to every underlying in the sample. This design choice is intended to limit overfitting and make the cross-asset comparison more meaningful.

4 Results

4.1 Performance on SPY

4.1.1 Return summary

Table 4: SPY performance summary under updated parameter settings.

Agent	Win Rate	Sharpe	Sortino	Ann. Return	Ann. Vol.	Max DD	Calmar	Kelly
Agent_hardThreshold	<u>75.00%</u>	<u>1.3248</u>	<u>1.1919</u>	<u>25.96%</u>	17.42%	<u>-11.12%</u>	<u>2.3339</u>	<u>7.6042</u>
Agent_Perc	74.36%	1.2669	1.1843	24.85%	17.52%	<u>-11.12%</u>	2.2345	7.2312
Agent_Garch	74.47%	1.0765	0.9319	21.49%	18.09%	-14.99%	1.4338	5.9520
Agent_LongTerm	73.91%	1.0618	1.1015	25.56%	<u>21.44%</u>	-11.95%	2.1381	4.9534

Table 4 summarizes risk-adjusted performance. **Agent_hardThreshold** delivers the strongest Sharpe ratio (1.3248) and Calmar ratio (2.3339). **Agent_Perc** is close in both return and risk metrics. **Agent_LongTerm** achieves high annual return (25.56%) with higher volatility (21.44%), while **Agent_Garch** remains profitable but with lower risk-adjusted efficiency in this run.

Figure 4 reports cumulative value-ratio paths (top panel) and daily return series (bottom panel) for SPY and the four strategy agents under the updated parameter set. All four agents outperform the underlying over the sample, with final value ratios clustered around 1.45–1.57 versus roughly 1.30 for SPY.

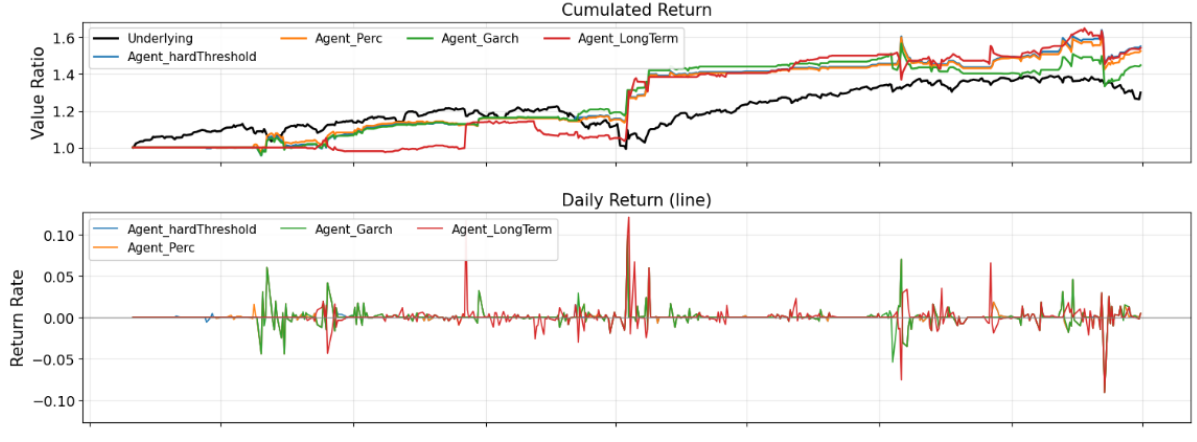


Figure 4: SPY cumulative return (value ratio) and daily return comparison across agents.

4.1.2 P&L attribution

Daily P&L is decomposed via a second-order Greek expansion:

$$dV \approx \underbrace{\Delta dS}_{\text{Delta}} + \underbrace{\frac{1}{2}\Gamma dS^2}_{\text{Gamma}} + \underbrace{\nu d\sigma}_{\text{Vega}} + \underbrace{\Theta dt}_{\text{Theta}} + \underbrace{\rho dr}_{\text{Rho}} + \underbrace{\text{Vanna } dS d\sigma}_{\text{Vanna}} + \underbrace{\frac{1}{2}\text{Volga } (d\sigma)^2}_{\text{Volga}} + \varepsilon.$$

The P&L from delta re-hedging trades is merged into the Gamma term, so the Gamma attribution captures both the convexity gain and the realized cost or benefit of maintaining the hedge.

For each Greek component i , the reported percentage contribution is computed as

$$\text{Greek } \%_i = \frac{\text{Greek}_i \text{ summed attribution}}{\sum_j |\text{Greek}_j \text{ summed attribution}|} \times 100.$$

Figure 5 shows the summed attribution breakdown for each agent. Vega is the dominant P&L source across all strategies, confirming that returns are primarily driven by volatility mispricing rather than directional exposure. Notably, **Agent_LongTerm** records the highest vega attribution (58.2%), indicating that the long-horizon realized-volatility signal is the most effective at capturing volatility mispricing and benefiting from implied volatility mean-reversion.

Summed Greeks Attribution Breakdown

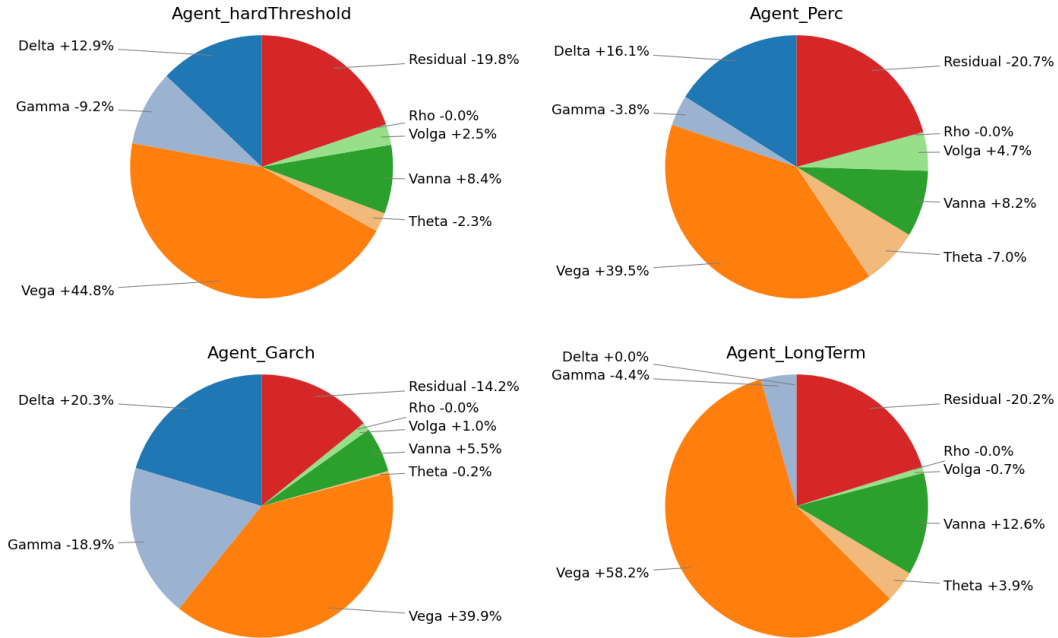


Figure 5: Summed Greeks attribution breakdown by strategy agent on SPY.

4.2 Performance on other ETFs

Table 5 reports performance statistics for QQQ and DIA under the same parameter set used in the SPY section. DIA is stronger on a risk-adjusted basis across agents (higher Sharpe and Sortino), while QQQ remains profitable but with larger drawdowns and more dispersion across agents.

Table 5: QQQ and DIA performance summary by agent.

Ticker	Agent	Win Rate	Sharpe	Sortino	Ann. Return	Ann. Vol.	Max DD
QQQ	Agent_hardThreshold	<u>81.08%</u>	0.8870	0.6754	21.10%	21.58%	-20.95%
QQQ	Agent_Perc	69.23%	0.9749	1.2191	34.63%	<u>30.50%</u>	-22.95%
QQQ	Agent_Garch	76.32%	1.1106	0.9158	28.83%	22.81%	-19.81%
QQQ	Agent_LongTerm	80.00%	<u>1.7709</u>	<u>1.5282</u>	<u>46.30%</u>	21.48%	<u>-14.83%</u>
DIA	Agent_hardThreshold	<u>87.50%</u>	1.5611	1.2112	47.21%	24.77%	-20.59%
DIA	Agent_Perc	87.04%	<u>2.2303</u>	<u>2.2453</u>	<u>91.18%</u>	29.06%	<u>-18.77%</u>
DIA	Agent_Garch	85.45%	1.9131	1.7752	68.01%	27.12%	-18.97%
DIA	Agent_LongTerm	85.71%	2.0476	1.7590	81.61%	<u>29.14%</u>	-19.89%

Table 6 reports summed Greek attribution percentages. Across QQQ and DIA, Vega is the largest positive contributor in every agent, consistent with a volatility-risk-premium strategy. The strongest Vega attribution appears in Agent_LongTerm for both ETFs (QQQ: +57.72%, DIA: +66.42%), indicating that this signal captures volatility-premium exposure most directly in the ETF sample. Residual attribution is generally negative, indicating that a non-trivial share of realized P&L is outside the low-order Greek decomposition.

Table 6: QQQ and DIA Greeks attribution percentages by agent.

Ticker	Agent	Delta	Gamma	Vega	Theta	Vanna	Volga	Rho	Residual
QQQ	Agent_hardThreshold	+2.81%	-12.76%	+54.67%	-0.76%	+2.73%	+9.64%	+0.00%	-16.62%
QQQ	Agent_Perc	<u>+10.38%</u>	<u>+9.51%</u>	+19.51%	-12.03%	<u>+6.83%</u>	<u>+15.87%</u>	-0.00%	-25.88%
QQQ	Agent_Garch	-5.33%	+8.92%	+47.41%	-6.11%	+2.40%	+9.83%	+0.01%	-19.98%
QQQ	Agent_LongTerm	-0.77%	-19.46%	<u>+57.72%</u>	<u>+3.42%</u>	+2.34%	+7.13%	<u>+0.02%</u>	<u>-9.14%</u>
DIA	Agent_hardThreshold	-4.10%	+3.97%	+48.70%	-10.06%	+6.17%	+14.41%	+0.00%	-12.58%
DIA	Agent_Perc	<u>-3.75%</u>	+2.60%	+50.89%	-8.11%	+4.56%	<u>+16.20%</u>	-0.00%	-13.88%
DIA	Agent_Garch	-8.63%	<u>+6.18%</u>	+58.07%	-8.56%	+2.44%	+9.69%	+0.00%	-6.43%
DIA	Agent_LongTerm	-6.71%	+1.86%	<u>+66.42%</u>	<u>+0.11%</u>	+1.02%	+7.41%	<u>+0.01%</u>	-16.46%

4.3 Performance on single stocks

Single-stock results are more heterogeneous than ETF results, reflecting larger idiosyncratic risk and event sensitivity. In terms of risk-adjusted performance, outcomes differ substantially by ticker: for example, GOOG and INTC are strong across multiple agents, whereas AMD and parts of ACMR/META are weaker or unstable in this sample.

From a Greeks-attribution perspective, Vega remains the dominant positive component in most ticker-agent combinations. The highest Vega attribution in the single-stock panel is **Agent_Garch** on ACMR (+67.53%), while **Agent_LongTerm** also shows very high Vega share on INTC (+42.03%) and TSLA (+41.71%). This supports the same core interpretation as in ETFs: realized strategy P&L is primarily tied to volatility exposure, with residual and cross-greek terms explaining the remainder.

The full single-stock statistics and Greeks-attribution tables are reported in Appendix A.

4.4 Robustness

The cross-asset results indicate that the long-horizon realized-volatility spread is the most robust signal in this study. Using one fixed parameter set (without ticker-specific tuning), it delivers the most consistent risk-adjusted performance across SPY, QQQ, and DIA, with mixed but still competitive outcomes in several single stocks.

Greeks attribution is aligned with this conclusion: Vega is the dominant positive contributor in most cases, indicating that returns are primarily coming from volatility mispricing rather than directional beta.

Robustness is stronger in ETFs than in single names, likely because ETFs have better liquidity and lower idiosyncratic jump risk. Main limitations are sample length, execution/transaction-cost assumptions, and residual terms outside the low-order Greeks decomposition.

5 Discussion

References

- [1] Bossu, Sébastien (2014). *Advanced Equity Derivatives: Volatility and Correlation*. Wiley Finance Series, John Wiley & Sons, Inc., Hoboken, NJ.

A Full Single-Stock Tables

Table 7 and Table 8 provide the complete single-stock results used to summarize this subsection.

Table 7: Full single-stock performance statistics by ticker and agent.

Ticker	Agent	Win Rate	Sharpe	Sortino	Ann. Return	Ann. Vol.	Max DD
AAPL	Agent_hardThreshold	73.17%	0.3969	0.2537	8.21%	19.88%	-21.25%
AAPL	Agent_Perc	72.22%	0.1404	0.0941	2.95%	<u>20.74%</u>	-26.44%
AAPL	Agent_Garch	73.33%	0.1402	0.0891	2.87%	20.16%	-23.40%
AAPL	Agent_LongTerm	<u>86.67%</u>	<u>0.5435</u>	<u>0.3926</u>	<u>9.08%</u>	15.99%	-16.86%
ACMR	Agent_hardThreshold	55.56%	-0.0069	-0.0065	-0.27%	39.16%	-25.42%
ACMR	Agent_Perc	61.54%	0.5744	<u>0.5040</u>	58.52%	<u>80.20%</u>	-32.10%
ACMR	Agent_Garch	60.00%	0.4892	0.4964	24.98%	45.58%	-31.94%
ACMR	Agent_LongTerm	60.00%	-0.9640	-0.4432	-48.66%	69.16%	-42.17%
AMD	Agent_hardThreshold	71.43%	-0.4379	-0.2366	-26.84%	<u>71.39%</u>	-65.30%
AMD	Agent_Perc	64.71%	-0.4454	-0.2423	-27.12%	71.04%	-66.78%
AMD	Agent_Garch	71.11%	-0.4153	-0.2214	-25.64%	71.35%	-63.65%
AMD	Agent_LongTerm	<u>85.71%</u>	<u>-0.0922</u>	<u>-0.0424</u>	<u>-6.12%</u>	68.56%	-59.33%
GOOG	Agent_hardThreshold	79.49%	<u>1.6039</u>	1.9659	58.84%	28.85%	-11.57%
GOOG	Agent_Perc	<u>81.58%</u>	1.1861	1.3496	50.25%	<u>34.32%</u>	-16.75%
GOOG	Agent_Garch	<u>72.09%</u>	1.5513	<u>2.0530</u>	<u>62.73%</u>	31.39%	-11.57%
GOOG	Agent_LongTerm	76.32%	0.9161	1.0031	32.45%	30.68%	-22.61%
INTC	Agent_hardThreshold	77.50%	1.4840	2.3184	124.20%	54.40%	-29.63%
INTC	Agent_Perc	71.05%	1.3589	<u>2.8220</u>	108.55%	54.09%	-24.79%
INTC	Agent_Garch	72.00%	<u>1.4840</u>	2.3462	<u>129.95%</u>	<u>56.11%</u>	-27.41%
INTC	Agent_LongTerm	76.09%	1.0758	0.9528	47.21%	35.94%	-28.73%
META	Agent_hardThreshold	65.79%	-0.1857	-0.1634	-5.96%	33.10%	-32.14%
META	Agent_Perc	65.00%	-0.3303	-0.2649	-9.86%	31.41%	-36.48%
META	Agent_Garch	<u>65.79%</u>	<u>0.1389</u>	<u>0.1267</u>	<u>4.79%</u>	<u>33.72%</u>	-28.72%
META	Agent_LongTerm	58.33%	-0.0927	-0.0777	-2.67%	29.20%	-24.03%
TSLA	Agent_hardThreshold	68.89%	<u>0.7805</u>	<u>0.7674</u>	<u>29.31%</u>	<u>32.93%</u>	-25.02%
TSLA	Agent_Perc	63.64%	0.5477	0.5530	18.19%	30.51%	-27.21%
TSLA	Agent_Garch	71.43%	0.7275	0.7003	23.56%	29.08%	-23.68%
TSLA	Agent_LongTerm	<u>72.34%</u>	0.1001	0.0701	2.23%	21.99%	-20.69%

Table 8: Full single-stock Greeks attribution percentages by ticker and agent.

Ticker	Agent	Delta	Gamma	Vega	Theta	Vanna	Volga	Rho	Residual
AAPL	Agent_hardThreshold	+17.52%	-30.20%	+26.97%	<u>+10.41%</u>	+3.20%	-1.70%	-0.01%	-9.99%
AAPL	Agent_Perc	+9.52%	-11.16%	+21.49%	<u>+5.86%</u>	<u>+4.14%</u>	<u>+15.59%</u>	-0.01%	-32.22%
AAPL	Agent_Garch	+9.79%	-21.20%	+28.28%	+8.64%	+3.17%	+7.35%	-0.00%	-21.56%
AAPL	Agent_LongTerm	+10.77%	-8.78%	<u>+46.82%</u>	-1.95%	+1.41%	-1.72%	<u>+0.00%</u>	-28.56%
ACMR	Agent_hardThreshold	+6.44%	-7.34%	+55.56%	-8.52%	+1.03%	<u>+1.92%</u>	+0.00%	-19.19%
ACMR	Agent_Perc	<u>+23.47%</u>	-6.99%	+48.71%	<u>-6.22%</u>	-6.62%	-4.91%	+0.02%	-3.07%
ACMR	Agent_Garch	-7.58%	+6.03%	<u>+67.53%</u>	-6.54%	+1.89%	+1.11%	+0.02%	-9.31%
ACMR	Agent_LongTerm	-1.19%	<u>+36.75%</u>	+5.91%	-15.02%	<u>+10.63%</u>	-9.47%	<u>+0.06%</u>	-20.97%
AMD	Agent_hardThreshold	+7.15%	-21.53%	+8.63%	<u>+8.58%</u>	+3.72%	+25.47%	+0.00%	-24.91%
AMD	Agent_Perc	+6.02%	<u>-13.35%</u>	+8.19%	+4.76%	<u>+5.05%</u>	<u>+30.28%</u>	<u>-0.00%</u>	-32.35%
AMD	Agent_Garch	<u>+8.88%</u>	-22.23%	+13.71%	+7.24%	<u>+4.27%</u>	<u>+20.53%</u>	<u>+0.00%</u>	-23.15%
AMD	Agent_LongTerm	-5.72%	-13.49%	<u>+50.14%</u>	+5.77%	-1.82%	+7.03%	<u>+0.00%</u>	-16.03%
GOOG	Agent_hardThreshold	+12.85%	-18.27%	+21.75%	<u>+13.61%</u>	-1.79%	+11.25%	-0.01%	-20.46%
GOOG	Agent_Perc	+11.50%	<u>-2.96%</u>	+24.81%	+2.14%	-1.25%	<u>+20.70%</u>	<u>-0.01%</u>	-36.61%
GOOG	Agent_Garch	+14.37%	-15.67%	+19.82%	+9.32%	+0.16%	<u>+14.76%</u>	<u>-0.01%</u>	-25.89%
GOOG	Agent_LongTerm	<u>+17.89%</u>	-2.98%	<u>+26.98%</u>	+3.76%	<u>+0.17%</u>	+11.49%	<u>-0.01%</u>	-36.71%
INTC	Agent_hardThreshold	-5.17%	+7.87%	+20.40%	<u>+10.16%</u>	+2.78%	+20.65%	<u>-0.00%</u>	-32.97%
INTC	Agent_Perc	<u>+5.30%</u>	<u>+11.22%</u>	+14.44%	-1.52%	+3.31%	<u>+24.23%</u>	<u>-0.00%</u>	-39.98%
INTC	Agent_Garch	+0.85%	-0.51%	+31.48%	+8.82%	+4.69%	+16.82%	<u>-0.00%</u>	-36.82%

Ticker	Agent	Delta	Gamma	Vega	Theta	Vanna	Volga	Rho	Residual
INTC	Agent_LongTerm	-10.42%	+10.34%	<u>+42.03%</u>	+3.15%	<u>+5.71%</u>	+3.51%	<u>-0.00%</u>	<u>-24.83%</u>
META	Agent_hardThreshold	-19.06%	-24.77%	+9.40%	+27.99%	+0.25%	-3.99%	+0.02%	+14.53%
META	Agent_Perc	-26.31%	<u>-20.21%</u>	<u>+26.12%</u>	+9.76%	+0.01%	<u>+16.33%</u>	<u>+0.02%</u>	-1.24%
META	Agent_Garch	-17.56%	<u>-27.06%</u>	+9.44%	<u>+33.54%</u>	-0.02%	+0.73%	+0.01%	+11.65%
META	Agent_LongTerm	<u>+6.50%</u>	-48.47%	+10.06%	<u>+22.45%</u>	<u>+2.51%</u>	+2.28%	+0.00%	+7.72%
TSLA	Agent_hardThreshold	<u>+5.24%</u>	-13.41%	+50.83%	<u>+2.63%</u>	+2.89%	-24.38%	+0.00%	<u>+0.61%</u>
TSLA	Agent_Perc	+1.06%	<u>+10.34%</u>	+32.66%	<u>-10.97%</u>	+2.58%	<u>+8.57%</u>	<u>+0.00%</u>	-33.79%
TSLA	Agent_Garch	-2.72%	+1.94%	<u>+53.48%</u>	-4.59%	+3.01%	+1.40%	<u>+0.00%</u>	-32.85%
TSLA	Agent_LongTerm	-11.85%	+5.65%	+41.71%	-3.36%	<u>+3.69%</u>	+5.16%	<u>-0.00%</u>	-28.58%

B Greeks Attribution: Dollar P&L Time Series

Figure 6 shows the daily dollar P&L contribution of each Greek component across the full backtest period. Each panel corresponds to one strategy agent and breaks down the daily total P&L into delta, gamma, vega, theta, vanna, volga, rho, and residual buckets. The time series view complements the pie chart by revealing when and for how long each risk factor dominated, and where attribution spikes occurred.

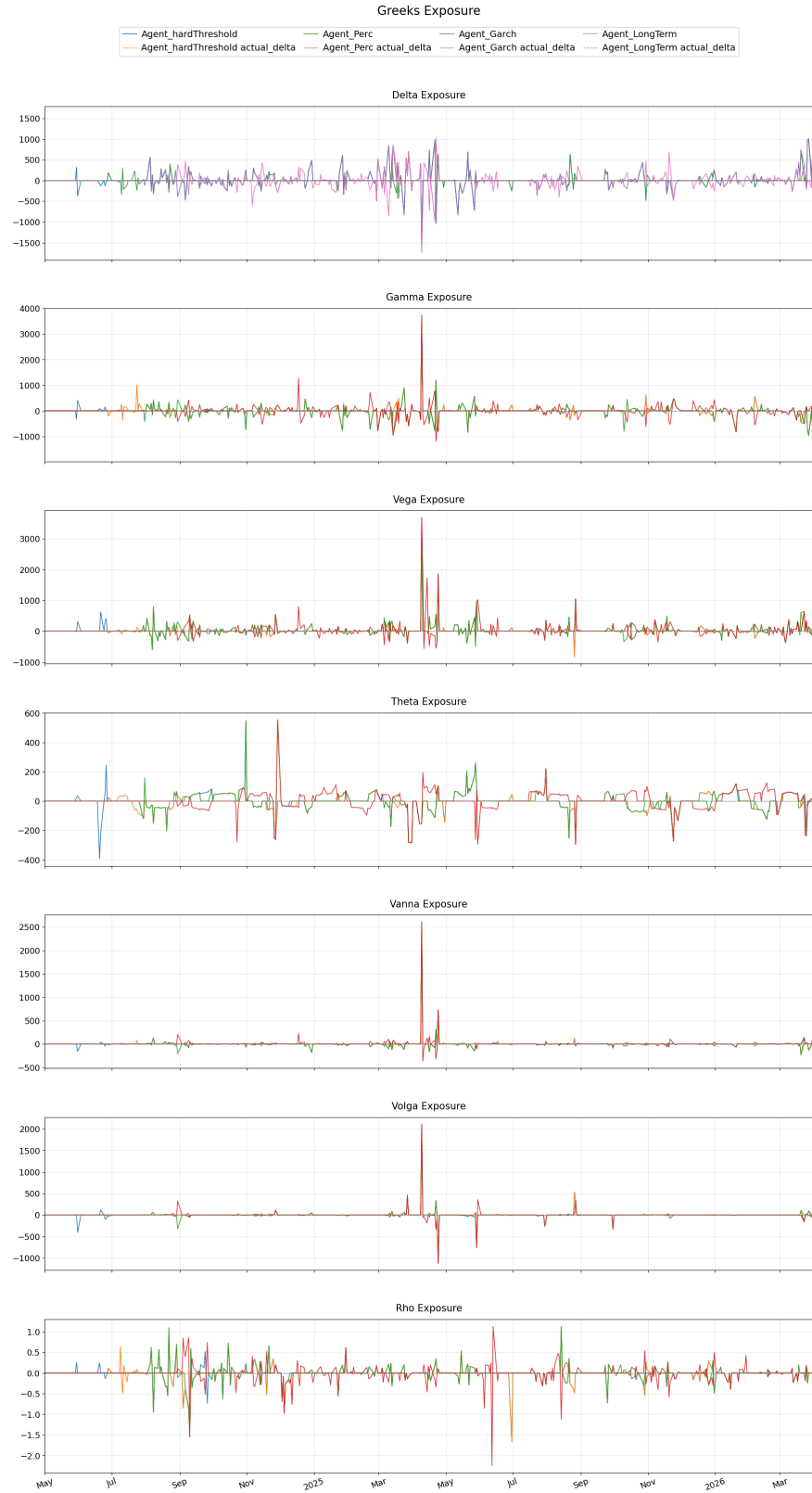


Figure 6: Daily dollar P&L attribution by Greek component for each agent on SPY.

C Greeks Exposure Over Time

Figure 7 shows the evolution of Greeks exposure over the backtest period. Unlike the dollar attribution series, this plot tracks the raw sensitivity levels (e.g. net vega, net delta, net gamma) as the position changes, providing insight into how risk exposures build up and unwind across entry, holding, and exit phases.

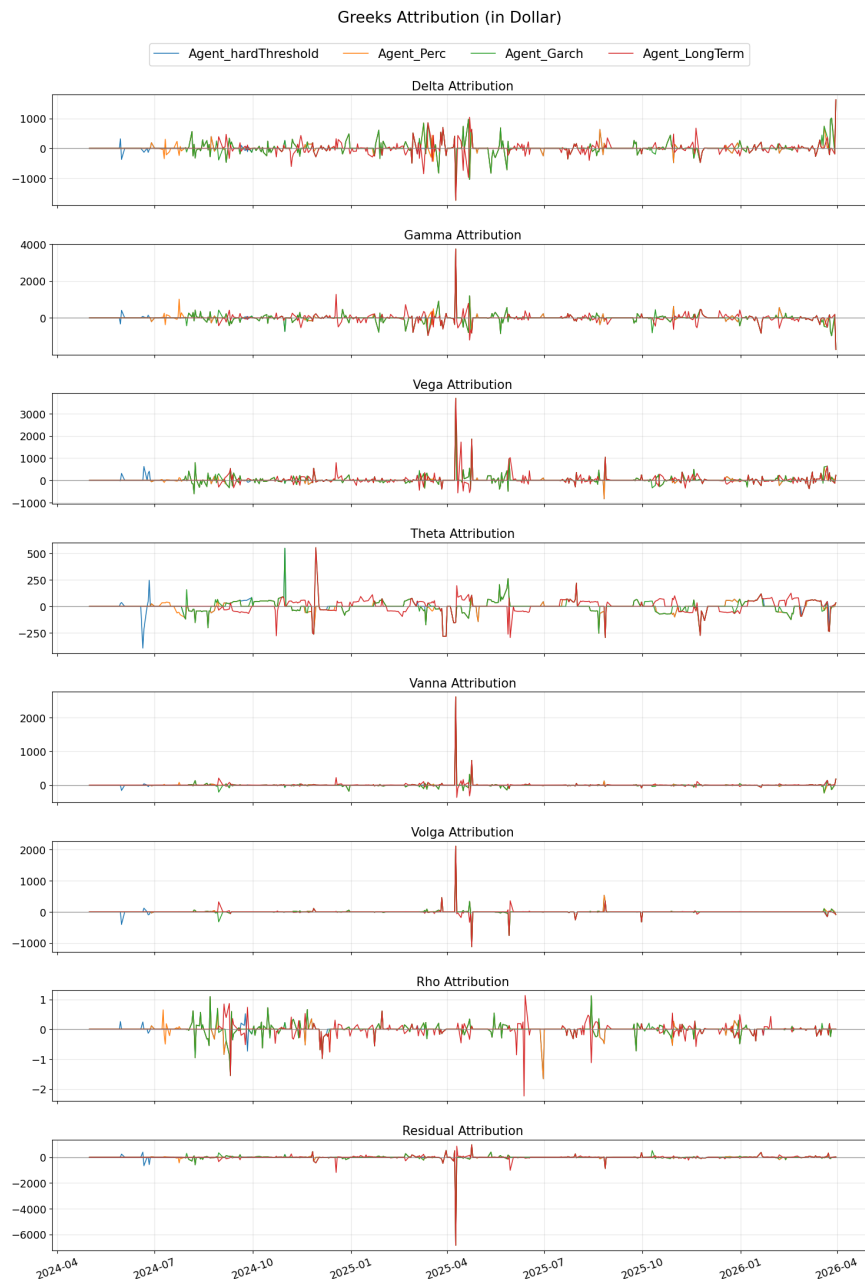


Figure 7: Greeks exposure over time for each agent on SPY.

D QQQ: Return and Greeks

Figure 8 and Figure 9 report the return and Greeks diagnostics for QQQ.

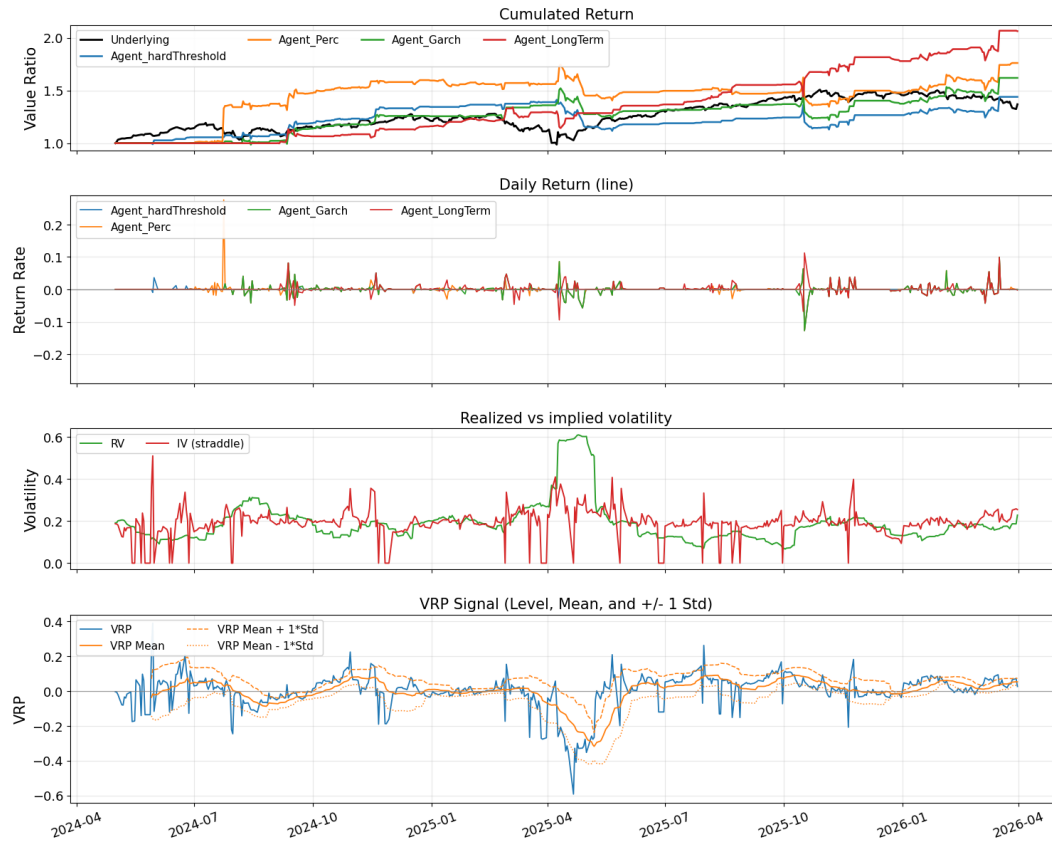


Figure 8: QQQ return diagnostics.

Summed Greeks Attribution Breakdown

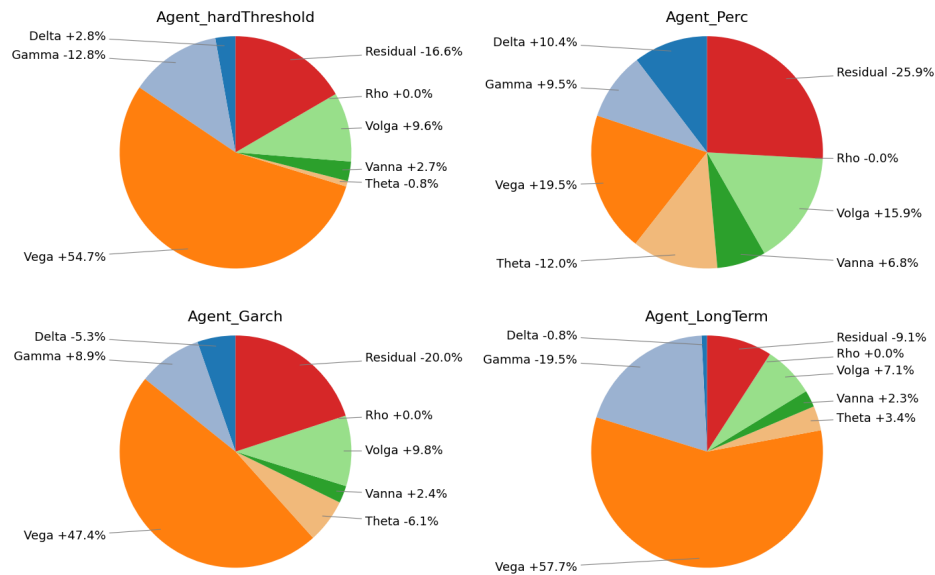


Figure 9: QQQ Greeks diagnostics.

E DIA: Return and Greeks

Figure 10 and Figure 11 report the return and Greeks diagnostics for DIA.



Figure 10: DIA return diagnostics.

Summed Greeks Attribution Breakdown

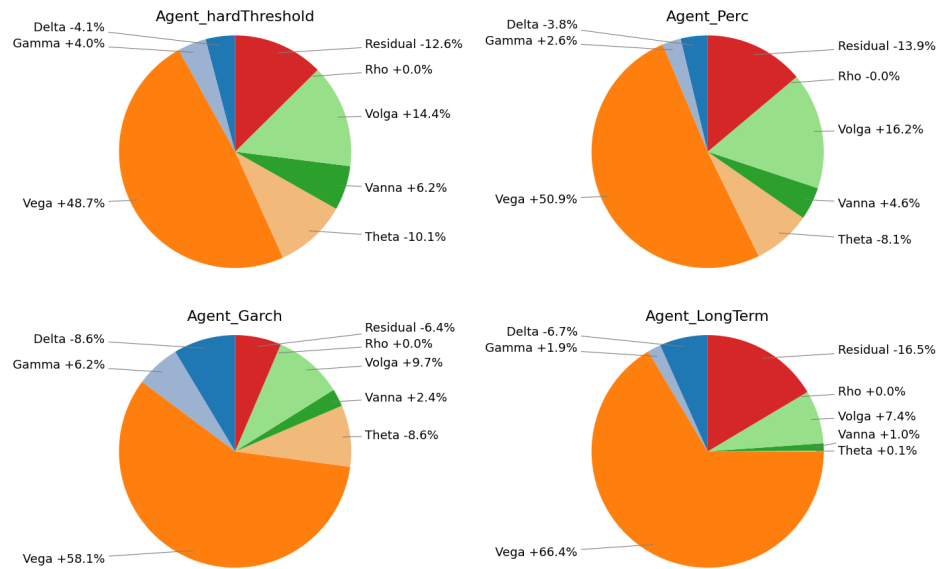


Figure 11: DIA Greeks diagnostics.